ShadowWeave: A Calibrated World Model of Occluded Space from a Single Depth Frame

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Abstract

An embodied agent has to reason about the parts of a scene it cannot directly see: the occupancy hidden by the walls and obstacles it is moving among (the "shadow"). A world model that represents only the visible surface is blind to half of its workspace. We ask how well a model can **complete** occupancy in that hidden region from a **single monocular-depth frame**, and, just as importantly, whether it knows *when it is guessing*. ShadowWeave projects one depth frame into an egocentric bird's-eye-view (BEV) occupancy grid, marks the unobserved cells with an explicit shadow mask, and predicts occupancy inside that mask together with a per-cell uncertainty. On a MuJoCo indoor benchmark, the model beats a persistence baseline by a micro-averaged shadow-IoU gain of **+0.31 to +0.36 at a 5 s horizon** (95% per-episode bootstrap CIs clear of zero at every horizon), far exceeds an observation-blind dataset prior even at the prior's best threshold (0.61–0.72 IoU vs 0.08–0.10), and its uncertainty is **better calibrated inside shadow (ECE 0.043–0.047, at 5 s) than in directly observed space (0.058–0.086)**. We frame prediction into occluded space as **world modeling of the unobserved physical scene**, inferring the occupancy state a downstream controller needs where the sensor has no view, and we introduce a decomposition that separates *amodal completion* of persistent hidden structure from *temporal forecasting* of dynamics. This gives an honest picture of the single-frame world model: its predictive power is almost entirely spatial completion, and the pure-forecasting residual is small ($\leq$+0.007 at 5 s) and decays with horizon. Randomizing room geometry per episode leaves the gain statistically unchanged on rooms the model never saw, which rules out memorization of a fixed room shell. The completion path runs at 7 ms (p95), well inside a 20 Hz control budget, and end-to-end with an off-the-shelf monocular-depth network it runs at $\sim$18 Hz on 2017-era hardware. The completed grid and its uncertainty feed an eyes-free spatial-audio navigation interface, which we include as a systems demonstration. Finally, we release the evaluation methodology, namely an empty-credit-immune micro-averaged gain, a completion/forecasting decomposition, and per-episode confidence intervals, as a reusable protocol for honest occupancy-completion claims.

1 Introduction

Any embodied agent acting in the physical world has to reason about the parts of a scene it cannot directly observe: the occupancy hidden by the very walls and obstacles it is moving among. A world model that represents only what the sensor sees is blind to half of the workspace. A robot

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rounding a shelf, an agent entering a room, or a planner routing around a blind corner all depend on the *unobserved* physical state as much as on the visible one, and completing that hidden state is a prerequisite for reasoning about what happens next. Sensing gives only a partial view, so the rest has to be inferred, and for any downstream use, whether that is planning, control, or a human-facing interface, the inference is only trustworthy if it carries an honest estimate of its own reliability. A confident hallucination of clear space behind a wall is worse than a calibrated "I don't know."

Prior work addresses parts of this problem but not their conjunction. Semantic scene completion, from MonoScene [Cao and de Charette, 2022] through VoxFormer [Li et al., 2023] and VisHall3D [Lu et al., 2025], hallucinates dense geometry beyond the visible frontier from a single image, but it targets outdoor driving voxels scored by mIoU alone, with no notion of calibrated confidence in the hallucinated region. Occupancy world models such as OccWorld [Zheng et al., 2024] forecast the *temporal* evolution of multi-frame occupancy for autonomous driving. Closest to our setting, ProxMaP [Sharma et al., 2023] completes indoor proximal occupancy from a single top-down view to improve navigation, but it emits a point estimate, with no explicit occlusion mask, no calibrated uncertainty in the hidden cells, and no attribution of *where* its predictive power comes from.

We present **ShadowWeave**, a world model of occluded space. From a single monocular-depth frame it produces (i) an egocentric BEV occupancy grid, (ii) an explicit shadow mask marking the cells the sensor cannot see, and (iii) a calibrated per-cell occupancy probability inside that mask. Unlike occupancy world models built for temporal forecasting, it models the *unobserved present*, the state behind obstacles, and we quantify exactly how much of its predictive power is spatial completion rather than temporal forecasting. We evaluate it with a protocol designed to resist the standard failure modes of occupancy metrics: an empty-credit-immune, micro-averaged shadow-IoU gain over a persistence baseline; per-episode bootstrap confidence intervals; and a static/dynamic/never decomposition that attributes the gain to *completion* of persistent hidden structure versus *forecasting* of dynamics. Downstream, the completed grid and its uncertainty feed an A* planner and a nine-zone HRTF spatial-audio interface for eyes-free navigation, which we include as a systems demonstration.

Our contributions are:

- **A calibrated amodal-completion result.** From one depth frame, the model completes occluded occupancy with a micro-averaged shadow gain of +0.31–0.36 at 5 s (CIs clear of zero at every horizon), and its in-shadow uncertainty is *better* calibrated (ECE 0.043–0.047 at 5 s) than in observed space.
- **A decomposition methodology** that separates completion from forecasting. It reveals that the gain is spatial completion of persistent hidden structure; the pure-forecasting residual is at or near zero (fixed static +0.000; moving tier +0.007 [+0.002, +0.012] at 5 s, which is statistically significant but practically small, and decays with horizon as real forecasting should). We report this negative honestly, since most pooled occupancy numbers in the literature are vulnerable to exactly the confound this decomposition dissects.
- **A memorization confound-kill.** Randomizing room geometry per episode leaves the gain statistically unchanged on never-seen rooms (+0.306 fixed vs +0.309 [+0.294, +0.324] randomized), showing the completion is per-scene inference, not a memorized constant layout.
- **A real-time completion front-end.** The completion path runs at 7 ms p95 (well inside a 20 Hz budget), and end-to-end with an off-the-shelf monocular-depth network it is ∼18 Hz on 2017-era hardware. Its output feeds an A* + nine-zone HRTF eyes-free interface, which we present as a systems demonstration (it does not yet meet navigation collision targets; §4.7).

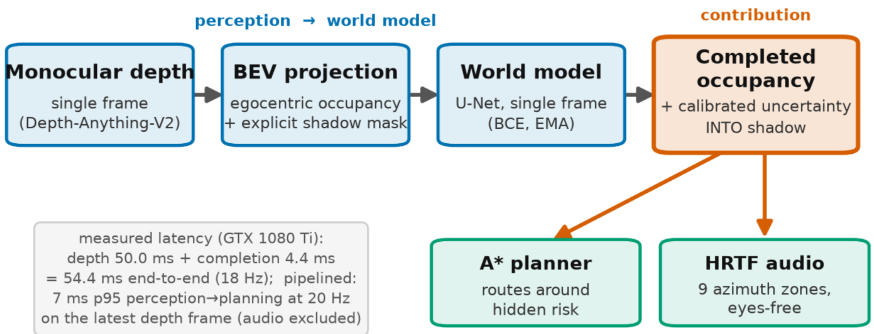

Figure 1: ShadowWeave pipeline. A single monocular-depth frame is lifted into an egocentric 96×96 BEV occupancy grid plus a visibility grid; thresholding visibility yields the shadow mask. A 31M-parameter U-Net completes per-cell occupancy at 1/3/5/10 s inside the mask; the completed grid and its uncertainty feed an A* planner and a nine-zone HRTF audio interface (system context).

2 Method

2.1 From a depth frame to an egocentric BEV with a shadow mask

At each step the agent observes a single monocular-depth image (in simulation, the renderer's depth; in deployment, a monocular-depth network such as Depth-Anything-V2-Small). A differentiable projector lifts the depth map into an egocentric bird's-eye-view (BEV) occupancy grid of size 96×96 covering 5 m ahead, together with a **visibility** grid: cells along a ray up to the first returned surface are marked observed-free, the surface cell observed-occupied, and everything beyond the surface is **shadow**, geometry the sensor cannot see. Thresholding visibility at 0.5 yields the binary shadow mask that defines the region of interest for every metric in this paper (Fig. 1). Optical flow between consecutive BEV frames provides an optional motion channel.

2.2 World model

We treat the network as a **world model of occluded space**: it infers the environment's occupancy state where the sensor has no view, rather than only labeling what is directly measured. As §4.3 quantifies, its predictive power concentrates in spatial completion of persistent hidden structure rather than in temporal forecasting of dynamics, which is a property of this single-frame model class that our decomposition makes measurable. A U-Net (31M parameters) takes the single-frame BEV stack and predicts occupancy at horizons of 1, 3, 5, and 10 s as per-cell probabilities. It is trained with binary cross-entropy (with the positive class up-weighted, since BEV targets are $\approx$7% occupied) on 400 training episodes, using early stopping and exponentially-averaged (EMA) weights; we report the epoch-8 checkpoint (validation loss 0.300, validation IoU@5s 0.665). Ablations (§4.6) show the visibility and flow channels are redundant, since the completion signal is carried by the occupancy channel alone, so the deployable model reduces to depth→occupancy→completion. The input is a **single frame**: the model has no multi-frame memory, which is what makes the completion-vs-forecasting distinction (§3.2) sharp.

2.3 Uncertainty in shadow

The model's per-cell occupancy probability *is* its uncertainty estimate; the claim we test in §4.5 is that this probability is **calibrated inside the shadow region specifically**, not merely grid-wide (where easy free space dominates). Calibrating uncertainty in the occluded region is the property that makes the output safe to consume downstream.

2.4 Downstream eyes-free interface (system context)

The completed occupancy grid and its uncertainty drive an A* global planner (which owns goal-seeking and pays an extra cost to route through unobserved cells) and a reactive local collision-avoidance policy. Occupancy and uncertainty in nine azimuth zones are rendered as HRTF-spatialized audio cues for eyes-free navigation. We describe this interface for context and report its latency and navigation behavior honestly (§4.7); the audio interface itself is a system demonstration, not a user-evaluated result.

3 Evaluation methodology

Occupancy metrics are easy to report and easy to inflate. Because our claim lives entirely inside a sparse, mostly-empty sub-region (shadow is $\sim$92% never-occupied), we adopt a protocol built to resist the specific ways such a claim can look better than it is. We consider this protocol a contribution in its own right.

3.1 The empty-credit trap and micro-averaging

The natural masked IoU scores an empty prediction against an empty masked region as a perfect 1.0 ("correctly predicted nothing here"). Since most shadow cells are legitimately empty, this inflates both the model and the baselines, and the inflation is not uniform across sub-masks. We therefore report **micro-averaged** shadow IoU: we pool integer intersection and union counts across all frames and divide once, so rare sub-masks cannot each contribute a free 1.0. Empty denominators report as undefined, never as 1.0. The defended quantity is the **gain** over a persistence baseline (the observed frame carried forward), not the absolute level.

3.2 Completion vs. forecasting decomposition

Within the shadow mask, we classify each cell by its occupancy pattern across the prediction horizons. This is valid because all horizons are rendered from the same issue pose, so static geometry projects to identical cells at every horizon:

- **STATIC**: occupied at *all* horizons, i.e. persistent hidden structure (walls, static obstacles, settled debris); $\approx$7% of shadow. Recall here measures **amodal completion**.
- **DYNAMIC**: occupied at *some but not all* horizons, i.e. transient-in-window structure (arrivals, departures, pass-throughs); $\leq$1.5% of shadow. IoU here measures genuine **forecasting** beyond persistence.
- **NEVER**: occupied at *no* horizon, i.e. free hidden space; $\approx$92% of shadow. This is the false-positive test.

One caveat follows from the sampling. Because the split keys on the four rendered horizons $\{1, 3, 5, 10\}$ s, dynamics faster than the 1 s grid, for example debris that is aloft and settles within the first second, are absorbed into STATIC, which biases the split toward completion. The moving tier, whose obstacles are in continuous motion across the whole window, is therefore the cleaner test of forecasting, and it still shows only a small dynamic gain (§4.3); a finer horizon grid is future work.

Reporting completion and forecasting separately is what lets us state honestly that the gain is completion, not clairvoyance, a distinction the pooled shadow-IoU hides.

3.3 Per-episode bootstrap confidence intervals

Frames within an episode share geometry and pose and are correlated, so resampling frames would understate uncertainty. We resample **episodes** with replacement ($B = 10{,}000$), recompute the pooled micro-gain per replicate, and take the paired model-minus-persistence difference within each replicate

Table 1: Micro-averaged shadow-IoU gain over the persistence baseline at the 5 s horizon on the fixed-geometry validation set, per tier, with 95% per-episode bootstrap confidence intervals.

tier (fixed val, @5 s)	micro shadow gain	95% CI
static	+0.306	[+0.288, +0.327]
moving	+0.363	[+0.349, +0.378]
debris	+0.315	[+0.281, +0.350]

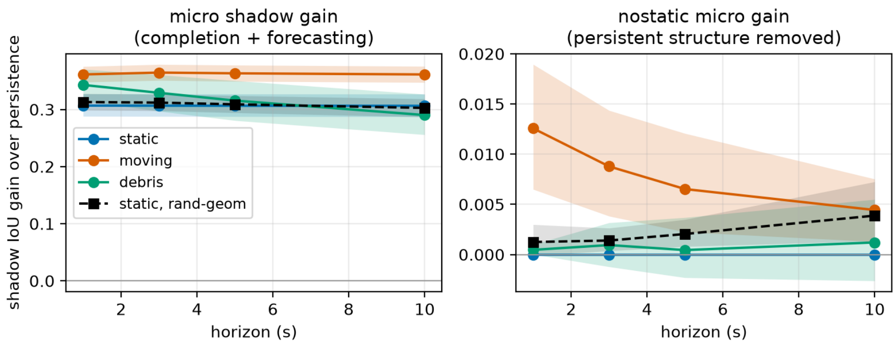

Figure 2: Micro-averaged shadow-IoU gain over persistence per tier and horizon, with 95% per-episode bootstrap confidence intervals. Every interval is clear of zero at every horizon, so the gain is statistically significant rather than sampling noise.

(the two are positively correlated across episodes). We report the resulting 2.5/97.5 percentile interval for every headline gain.

3.4 Calibration in shadow

We measure expected calibration error (ECE) restricted to shadow cells, contrasted with observed cells and with a walls-excluded variant, using equal-mass binning (a fine 1000-bin histogram merged into 15 equal-count macro-bins, because in-shadow probabilities cluster near the prior and equal-width bins would be mostly empty). An ECE-in-shadow that is close to or below ECE-in-observed is the evidence that uncertainty propagates faithfully into unseen space.

4 Experiments

4.1 Setup

We use a MuJoCo single-room benchmark with three difficulty tiers: **static** (fixed obstacles), **moving** (kinematically driven movers), and **debris** (objects that fall and settle). It has 400 training and 80 validation episodes on disjoint seeds; the per-tier decomposition and CI analyses use clean single-tier validation sets of 30 episodes (9,000 frames) each, and the randomized-geometry validation (§4.4) uses 80 episodes. The BEV is 96×96 over 5 m; horizons are 1/3/5/10 s. Unless noted, fixed-geometry results use a 6×6 m room; §4.4 randomizes geometry. Reproducibility is pinned: the fixed-geometry scene generation is verified bit-exact against a golden hash, and a full rollout re-run reproduces every accuracy metric bit-identically (wall-clock latency, which is hardware-dependent, is the one exception).

4.2 Shadow-gain headline

The model predicts occupancy in cells the sensor cannot see, beating persistence at every horizon. Micro-averaged shadow gain at 5 s, per tier, with 95% bootstrap CIs, is shown in Table 1.

Table 2: Completion/forecasting decomposition of the shadow mask on the fixed-geometry validation set at 5 s (dynamic IoU rows span the 1 s→10 s horizons). The gain is carried by recall of persistent hidden structure (amodal completion), not by forecasting of dynamics.

component (fixed val @5 s)	model	persistence
persistent-structure recall (amodal completion)	0.83–0.93	0.37–0.51
dynamic IoU, moving tier (1 s→10 s)	0.092 → 0.031	0.060 → 0.020
dynamic IoU, debris tier	0.020 → 0.019	0.006 → 0.023
false-positive rate on never-occupied cells	0.014–0.034	0.022–0.027

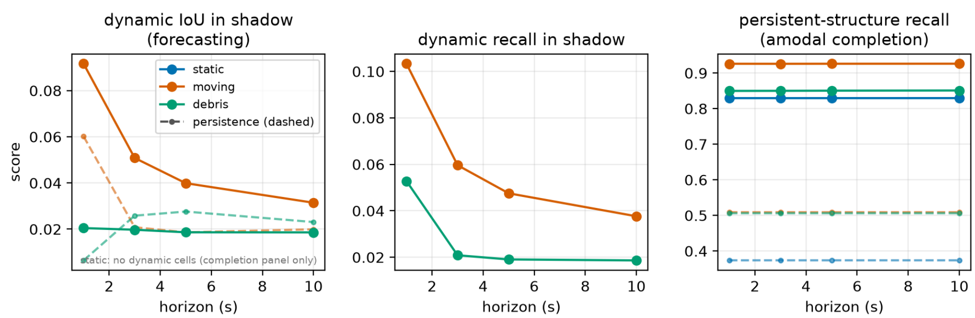

Figure 3: Decomposition of the shadow gain across horizons: persistent-structure recall (amodal completion) versus dynamic IoU (forecasting) for model and persistence. The gain is spatial completion of persistent hidden structure; the small dynamic-forecasting signal on the moving tier decays with horizon as genuine forecasting should.

Every interval is clear of zero at every horizon, so the gain is statistically significant rather than sampling noise (Fig. 2; all horizons tabulated in Appendix A). In policy rollouts the frame-averaged (macro) shadow gain is +0.325/+0.322/+0.320/+0.178 at 1/3/5/10 s. Raw macro shadow-IoU levels (model 0.744 vs persistence 0.424 at 5 s) are empty-credit inflated and are reported only for reference; the micro-gain is the defended number.

Beyond persistence. Persistence could be a weak baseline for *completion*, so we test two stronger ones (full tables in Appendix B). An observation-blind pose-marginal prior (mean training occupancy, swept to its best threshold) reaches only 0.08–0.10 micro shadow IoU: it recalls hidden structure (0.86–0.92) only by blanketing 72–79% of empty hidden space with false positives, whereas the model achieves 0.61–0.72 IoU at a 1.4–3.4% false-positive rate. An observation-conditioned dilation heuristic peaks at micro shadow IoU 0.31–0.35, at or below persistence itself. Hidden structure is thus non-local: copying the observation, extrapolating it locally, and an observation-blind prior all fail. Together with the geometry-randomization result (§4.4), this rules out memorized layouts, memorized marginals, and trivial extrapolation alike.

4.3 What the gain is: completion, not clairvoyance

Decomposing the shadow mask (§3.2) attributes the gain almost entirely to amodal completion of persistent hidden structure (Table 2, Fig. 3).

Excluding all persistent structure, the residual “nostatic” gain is at or near zero: fixed static +0.000, moving +0.007 (CI [+0.002, +0.012]), randomized-geometry static +0.002 ([+0.001, +0.004]) at 5 s. The honest claim is therefore **amodal completion of hidden structure**, with a small but statistically significant dynamic-forecasting signal on the moving tier that decays with horizon as genuine forecasting should. We do not claim dynamic forecasting on debris, where the dynamic signal is at the noise floor and persistence matches the model beyond 1 s.

Table 3: Static-tier shadow gain at 5 s under fixed versus per-episode randomized room geometry. The gain does not drop on never-seen rooms, ruling out memorization of a constant layout.

static tier, @5 s	micro shadow gain	95% CI	completion recall
fixed 6×6 geometry	+0.306	[+0.288, +0.327]	0.829
randomized geometry	**+0.309**	**[+0.294, +0.324]**	0.854

Table 4: Input-channel ablations. Shadow gain at 5 s (micro, per tier) and rollout shadow gain at 5 s for the full model and the no-visibility and no-flow variants.

variant	shadow gain @5 s (micro, static/moving/debris)	rollout shadow gain @5 s
full (occupancy+visibility+flow)	+0.306 / +0.363 / +0.315	+0.320
no visibility	+0.309 / +0.360 / +0.313	n/a
no flow	n/a	+0.381

4.4 Robustness to room geometry (memorization confound-kill)

A fixed room shell could let the model memorize a constant wall prior. We regenerated the data with per-episode randomized room width and depth (each drawn i.i.d. in $[5, 8]$ m, obstacle counts scaled to hold the $\sim$7% positive fraction), retrained the model from scratch, and evaluated on 80 held-out randomized rooms (Table 3).

The gain does not drop: the intervals overlap almost entirely, and completion recall is if anything slightly higher on never-seen rooms. Memorizing a constant shell contributes essentially nothing; completion is per-scene inference.

4.5 Calibration in shadow

Shadow is $\sim$92% empty, so a raw in-shadow ECE is easy to make small, because probability mass clusters near the prior. We therefore lead with the harder number. *Excluding the persistent structure entirely* (the walls and settled cells that dominate the mask), in-shadow ECE at 5 s remains a low 0.055–0.056 on the tiers with dynamic content (on the static tier the structure-excluded region has no positive labels, so no calibration curve exists there). Calibration is thus carried by the genuinely uncertain cells, not the easy always-occupied ones. On the full shadow mask at 5 s, in-shadow ECE is 0.043–0.047 across tiers, lower than in observed space (0.058–0.086), with Brier 0.022–0.036 (Fig. 4); we note that this shadow-vs-observed edge is aided by shadow's emptiness and is a secondary observation, not the primary claim. The model is underconfident in the mid-range but places little mass there. Calibration is not an artifact of the fixed room: on randomized geometry, in-shadow ECE is 0.050–0.067 vs observed 0.056–0.077, with shadow the better-calibrated region only at the longer 5–10 s horizons (at 1 s the two are comparable). The reliability diagram is in Appendix F (Fig. 4).

4.6 Ablations: occupancy alone carries the signal

Dropping either auxiliary input channel and retraining leaves the gain intact (Table 4).

The no-visibility per-tier gains are within CI of the full model; the no-flow rollout gain is if anything higher. Both auxiliary channels are redundant, which simplifies the input and removes optical-flow estimation from the runtime critical path. These independently retrained variants also bound training-run variance: every one of them (and the separately trained randomized-geometry model of §4.4) lands within the full model's per-tier confidence intervals, so the headline gain is not an artifact of a lucky training seed.

4.7 System: latency and navigation

Latency. The perception-to-planning path (BEV projection, raycaster, world-model forward, orchestrator; batch 1, deterministic U-Net) runs at 6.47 ms p50 / 6.99 ms p95, well inside a 50 ms (20 Hz) budget. Depth is the bottleneck: Depth-Anything-V2-Small alone takes 49.9 ms p50 on the evaluation-era GPU (GTX 1080 Ti, 480×640); a separate single-process benchmark of the sequential depth$\rightarrow$completion pipeline (whose in-process completion subtotal, without the orchestrator, is 4.4 ms) measures 54.4 ms end-to-end (18 Hz). A deployed system runs depth asynchronously at its own rate while the 7 ms completion path consumes the latest depth frame; audio synthesis is excluded.

We also probed whether the model flags occupancy that materializes in shadow before it is revealed. Because that metric conflates falling objects with static geometry revealed by agent motion and does not meet its lead-time target, we report it only as a diagnostic in Appendix C, not as a result or a contribution.

Navigation. With the completed grid feeding the planner, the trained reactive policy achieves a per-episode collision rate of 0.367 (vs 0.60 untrained) and path straightness 0.773, but does not meet the ≤ 0.10 collision target. We present navigation as an honest system demonstration.

A generative (conditional-diffusion) variant of the world model is deferred to Appendix D.

5 Related work

Predicting the structure of unobserved space has been approached from three angles. Semantic scene completion, from SSCNet [Song et al., 2017] through MonoScene [Cao and de Charette, 2022] and its monocular successors VoxFormer [Li et al., 2023] and VisHall3D [Lu et al., 2025], infers dense voxel geometry, and even hallucinates geometry beyond the visible frontier, from a single image, but it targets outdoor driving voxels evaluated purely by mIoU/IoU, without calibrated uncertainty in the occluded region. Occupancy world models such as OccWorld [Zheng et al., 2024], together with occupancy-forecasting benchmarks (Occ3D; Tian et al., 2023) and self-supervised raycasting forecasters [Khurana et al., 2023], instead predict the *temporal* evolution of multi-frame occupancy for autonomous vehicles. Closest in spirit, ProxMaP [Sharma et al., 2023] completes indoor proximal occupancy from a single view for navigation, yet emits only point estimates. ShadowWeave is distinct in delivering egocentric BEV amodal completion into explicitly masked occluded space from a single monocular-depth frame, with calibrated uncertainty inside the hidden region, and a completion-versus-forecasting decomposition showing the gain is spatial completion rather than dynamics.

6 Limitations

Our evaluation is in simulation on the simulator's clean depth: ground-truth occluded occupancy is unavailable on real footage without instrumentation, so no quantitative real-world claim is made and robustness of the completion to estimated-depth noise is untested, though the front-end accepts real monocular depth (Depth-Anything-V2). We compare against strong self-constructed baselines rather than prior methods, which target different input regimes (e.g., ProxMaP's top-down RGB-D view); a head-to-head on a shared benchmark is future work. Our results characterize a single 31M single-frame CNN, so the forecasting-negative applies to this model class, not necessarily to larger or recurrent ones (the evaluation protocol itself is architecture-agnostic). Navigation does not meet its collision target and the audio interface is not user-evaluated; both are system context.

7 Conclusion

Calibrated amodal completion of occluded occupancy is achievable from a single monocular-depth frame, transfers to room geometries the model never saw, and can be delivered within a real-time perception-to-planning budget for an eyes-free interface. Equally important is *how* we measure it: an empty-credit-immune micro-gain, per-episode confidence intervals, and a decomposition that separates completion from forecasting keep the claim honest and expose exactly where the predictive power lies. We hope the protocol is reused, since many occupancy-completion results would read differently under it.

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A Per-horizon shadow gains with confidence intervals

Table 5 reports the micro-averaged shadow-IoU gain over persistence at every horizon, with 95% per-episode bootstrap confidence intervals (10,000 resamples, paired model−persistence). Every interval is clear of zero. The static-tier gain is flat across horizons (persistent structure does not move); the debris-tier gain decays with horizon as more debris settles into persistent structure; the randomized-geometry run tracks the fixed-geometry static tier throughout.

B Heuristic baselines: observation-blind prior and dilation

The pose-marginal prior is the mean training-set occupancy grid in the egocentric frame, thresholded at whichever of eight thresholds (0.02–0.5) maximizes its micro shadow IoU, which is the most

Table 5: Micro shadow gain over persistence, all horizons, 95% bootstrap CIs over episodes.

tier	1 s	3 s	5 s	10 s
static	+0.307 [+0.288, +0.327]	+0.307 [+0.288, +0.327]	+0.306 [+0.288, +0.327]	+0.306 [+0.287, +0.327]
moving	+0.361 [+0.348, +0.375]	+0.364 [+0.351, +0.379]	+0.363 [+0.349, +0.378]	+0.361 [+0.347, +0.375]
debris	+0.343 [+0.315, +0.370]	+0.329 [+0.297, +0.361]	+0.315 [+0.281, +0.350]	+0.290 [+0.256, +0.327]
rand. geometry	+0.313 [+0.299, +0.328]	+0.312 [+0.298, +0.327]	+0.309 [+0.294, +0.324]	+0.303 [+0.287, +0.318]

favorable possible reading. Table 6 shows that even so, the prior can reach high structure recall only by blanketing ~72–79% of genuinely empty hidden space with false positives, capping its IoU at 0.08–0.10 versus the model's 0.61–0.72. One caution: the prior's IoU restricted to *dynamic* cells should never be quoted in isolation, because at its best threshold the prior carpet-bombs the dynamic region too, which is meaningless at a 72% false-positive rate.

Table 6: Observation-blind prior at its best threshold (0.05) vs the model, @5 s.

tier	prior micro IoU	prior structure recall	prior FP on empty	model (IoU @ FP)
static	0.102	0.879	0.723	0.610 @ 0.034
moving	0.080	0.865	0.722	0.723 @ 0.015
debris	0.089	0.869	0.717	0.670 @ 0.014
rand. geometry	0.087	0.921	0.793	0.655 @ 0.021

The observation-*conditioned* counterpart, binary dilation of the observed occupancy by $r \in \{1, \ldots, 6\}$ cells (one cell $\approx$10 cm), reported at the IoU-maximizing radius, probes whether local extrapolation of the observation suffices. It does not (Table 7): the heuristic peaks at r=1, essentially tied with or below plain persistence, because at matched false-positive rates it recalls only 0.42–0.56 of hidden structure versus the model's 0.83–0.93, and every larger radius grows false positives faster than recall, monotonically lowering IoU.

Table 7: Observation-conditioned dilation heuristic at its best radius (r=1) vs persistence and the model, micro shadow IoU @5 s.

tier	dilation (best r)	dilation recall @ FP	persistence	model
static	0.305	0.416 @ 0.035	0.303	0.610
moving	0.346	0.557 @ 0.039	0.360	0.723
debris	0.342	0.544 @ 0.039	0.355	0.670

C Occupancy-arrival diagnostic (not falling-object anticipation)

We keep this out of the main results because it does not measure what a "falling-object" metric should: events are counted per due prediction (one object aloft is scored at every issue step and horizon it is due, not once per physical object); an "arrival" keys on unobserved-at-issue rather than newly-occupied cells, so occluded *static* geometry revealed by agent motion is also counted, and the metric pools all tiers; and lead time is bounded by the discrete $\{1, 3, 5, 10\}$ s horizon grid, so the $\geq$3 s target is neither met nor meaningfully measurable at this granularity. We report it only to characterize the model's behavior, not as a capability claim.

The component-level variant splits each frame's arrival mask into connected components and scores each independently, which exposes what the cell-level false-alarm rate hides: of the distinct arrival components the model predicts, 29% are spurious. Cell-level detection is 0.787 at a horizon-granular mean lead of 2.62 s; the component-level scoring gives detection 0.752 and precision 0.714. The definitional caveats above apply to both columns; precision in particular is an unmatched majority-overlap over predicted components and is sensitive to fragmentation.

Table 8: Cell-level vs connected-component scoring of the revealed-occupancy proxy.

metric	cell-level	component-level
detection rate	0.787	0.752
false-alarm rate	0.021	0.286
precision	n/a	0.714
mean lead time (horizon-granular)	2.62 s	2.63 s

D Generative variant (ongoing work)

We additionally trained a conditional-diffusion (DDPM) world model as an alternative to the deterministic U-Net, motivated by the fact that BCE is mean-seeking and hedges into a grey smear exactly where the future is multimodal, which is the shadow, whereas a generative model can sample coherent alternatives and expose its disagreement as uncertainty concentrated in the occluded region. A full calibrated evaluation (in-shadow sample-diversity ratio and calibration, the fair comparison given that BCE and DDPM optimize different objectives) is ongoing: iterated DDIM sampling exceeds our batch-eval budget, and we leave a like-for-like generative comparison to future work. The deterministic model's calibrated per-cell probabilities already carry the uncertainty narrative of this paper.

E Reproducibility

Scene generation for the fixed-geometry environment is pinned by a golden-hash regression test (byte-identical XML for the first 50 seeds), and a full 90-episode rollout re-run reproduces every published accuracy metric bit-identically (wall-clock latency, which is hardware-dependent, is the one exception). All full-model numbers use a single early-stopped checkpoint selected on validation IoU; train, validation, and evaluation episodes use disjoint seed ranges. Code, configuration, seeds, the evaluation summaries behind every table, and a frozen dependency lock accompany the paper; datasets are regenerable from seeds with a single pipeline script.

F Reliability diagram

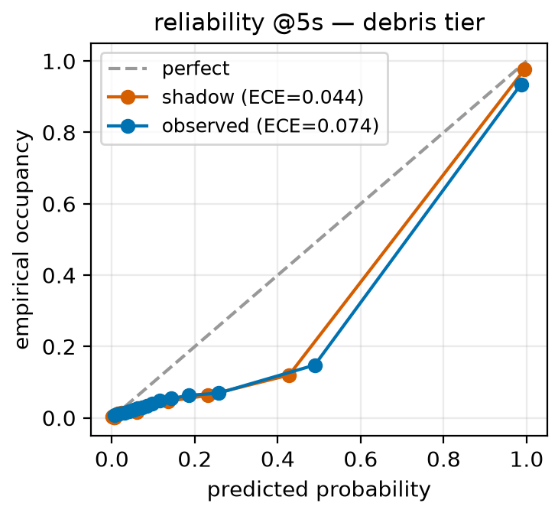

Figure 4: Reliability diagram restricted to shadow cells (equal-mass binning: a fine 1000-bin histogram merged into 15 equal-count macro-bins), debris tier. At 5 s, in-shadow ECE is 0.043–0.047 across tiers, lower than in directly observed space (0.058–0.086), with Brier 0.022–0.036; the model is underconfident in the mid-range but places little mass there.